

Unemployment rate prediction Using ARIMA

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2026-05-23

Project Overview

This project employs ARIMA (AutoRegressive Integrated Moving Average) modeling to analyze and forecast Kenya's unemployment rate trends from 2000 to 2029. Using annual time series data spanning 25 years (2000-2024), the study identifies underlying patterns and structural shifts in Kenya's labor market dynamics. The analysis is particularly relevant given the dramatic changes observed in recent years, including the sharp rise in unemployment following the 2017 period, potentially attributed to economic disruptions, policy transitions, and the global COVID-19 pandemic. By applying rigorous econometric techniques, this research provides evidence-based projections to inform policymakers, economists, and stakeholders in developing targeted employment strategies and fiscal interventions aimed at addressing Kenya's persistent unemployment challenges.

Setting working directory

```
setwd("D:/Analysis Projects")
```

Loading necessary libraries

```
library(readxl)      # For Excel files
library(tidyverse)    # Data manipulation

library(forecast)     # ARIMA modeling
library(tseries)      # Time series tests

library(ggplot2)      # Visualization
library(lubridate)    # Date handling
```

Data importation

```
data=read.csv("unemployment.csv")
head(data)
```

```
##   Year Unemployment_Rate
## 1 2000                2.89
## 2 2001                2.88
## 3 2002                2.92
## 4 2003                2.87
## 5 2004                2.85
## 6 2005                2.75
```

Data inspection

```
str(data)

## 'data.frame':    25 obs. of  2 variables:
##  $ Year          : int   2000 2001 2002 2003 2004 2005 2006 2007 2008
## 2009 ...
##  $ Unemployment_Rate: num   2.89 2.88 2.92 2.87 2.85 2.75 2.68 2.65 2.76
## 2.82 ...
```

Check missing values

```
colSums(is.na(data))

##           Year Unemployment_Rate
##           0              0
```

Create time series object

```
unemp_ts <- ts(data$Unemployment_Rate,
               start = 2000,
               end = 2024,
               frequency = 1)
```

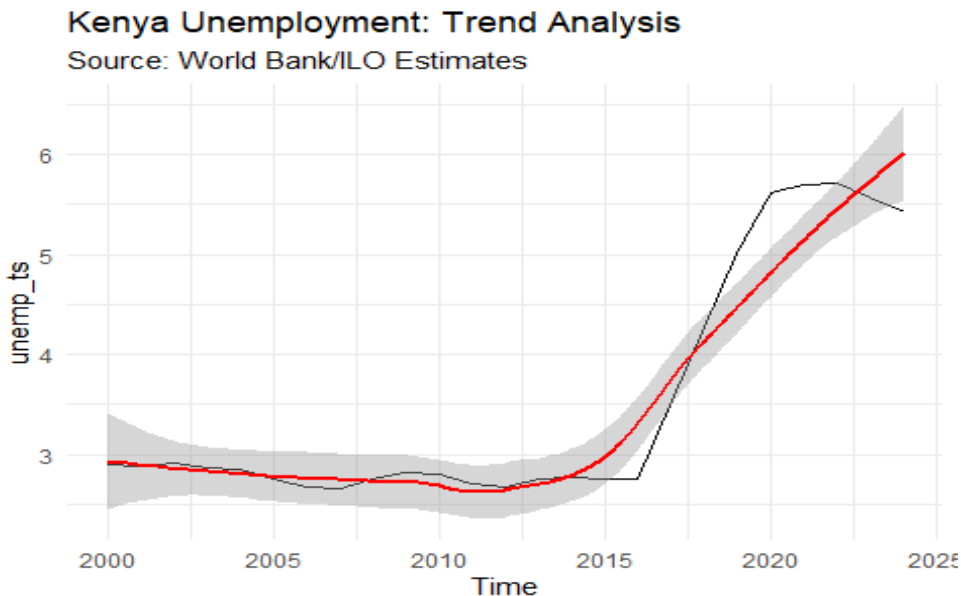
Visualize raw data

```
plot(unemp_ts,
     main = "Kenya Unemployment Rate (2000-2024)",
     xlab = "Year", ylab = "Unemployment Rate (%)",
     col = "steelblue", lwd = 2)
grid()
```



Exploratory Data Analysis & Stationarity Checks

```
autoplot(unemp_ts) +  
  geom_smooth(method = "loess", color = "red") +  
  labs(title = "Kenya Unemployment: Trend Analysis",  
        subtitle = "Source: World Bank/ILO Estimates") +  
  theme_minimal()  
  
## `geom_smooth()` using formula = 'y ~ x'
```



Stationarity Tests

ADF Test (Null: series has unit root = non-stationary)

```
adf_test <- adf.test(unemp_ts)  
print(adf_test)  
  
##  
## Augmented Dickey-Fuller Test  
##  
## data: unemp_ts  
## Dickey-Fuller = -2.4039, Lag order = 2, p-value = 0.4185  
## alternative hypothesis: stationary
```

KPSS Test (Null: series is stationary)

```
kpss_test <- kpss.test(unemp_ts)  
print(kpss_test)
```

```
##
## KPSS Test for Level Stationarity
##
## data: unemp_ts
## KPSS Level = 0.64405, Truncation lag parameter = 2, p-value = 0.01863

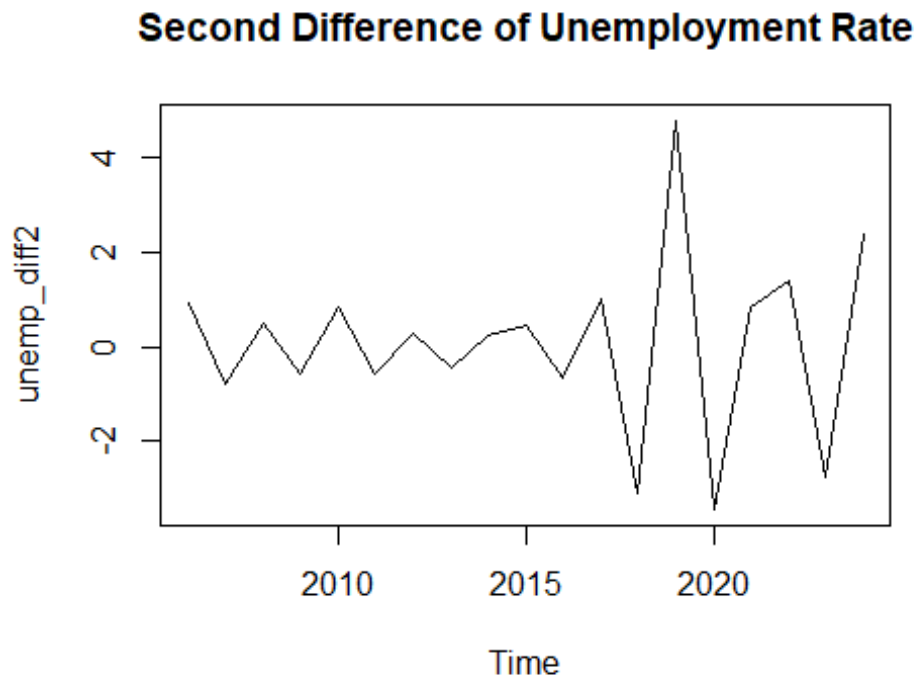
unemp_diff <- diff(unemp_ts, differences = 1)
unemp_diff1 <- diff(unemp_diff, differences = 2)
unemp_diff2 <- diff(unemp_diff1, differences = 3)
adf.test(unemp_diff2)

## Warning in adf.test(unemp_diff2): p-value smaller than printed p-value

##
## Augmented Dickey-Fuller Test
##
## data: unemp_diff2
## Dickey-Fuller = -4.5445, Lag order = 2, p-value = 0.01
## alternative hypothesis: stationary
```

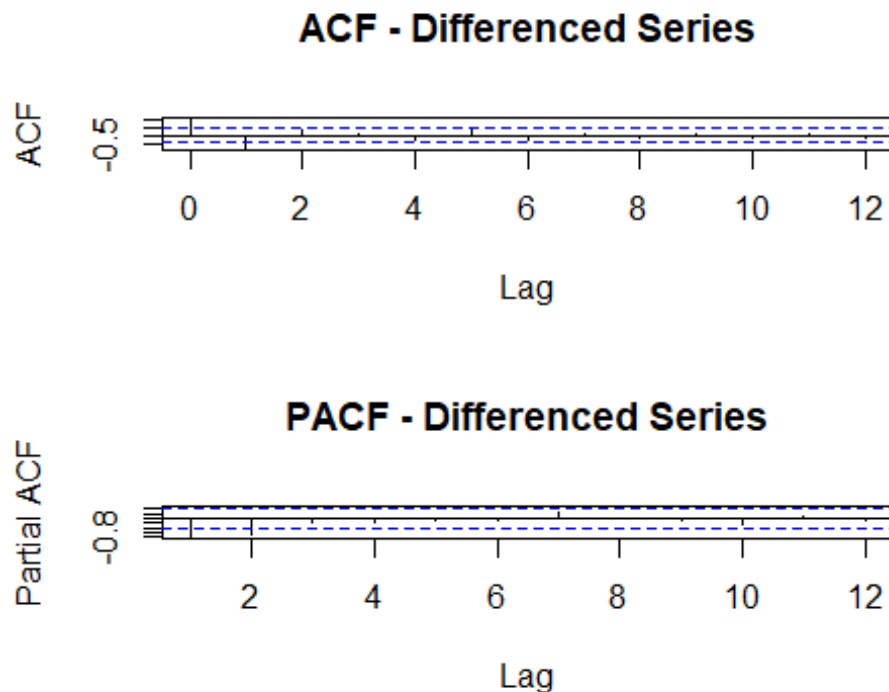
plot differenced rate

```
plot(unemp_diff2, main = "Second Difference of Unemployment Rate")
```



ACF & PACF plots for model identification

```
par(mfrow = c(2,1))
acf(unemp_diff2, main = "ACF - Differenced Series")
pacf(unemp_diff2, main = "PACF - Differenced Series")
```



Auto ARIMA (recommended starting point)

```
auto_model <- auto.arima(unemp_ts,
                          seasonal = FALSE, # Annual data
                          stepwise = FALSE,
                          approximation = FALSE)

summary(auto_model)

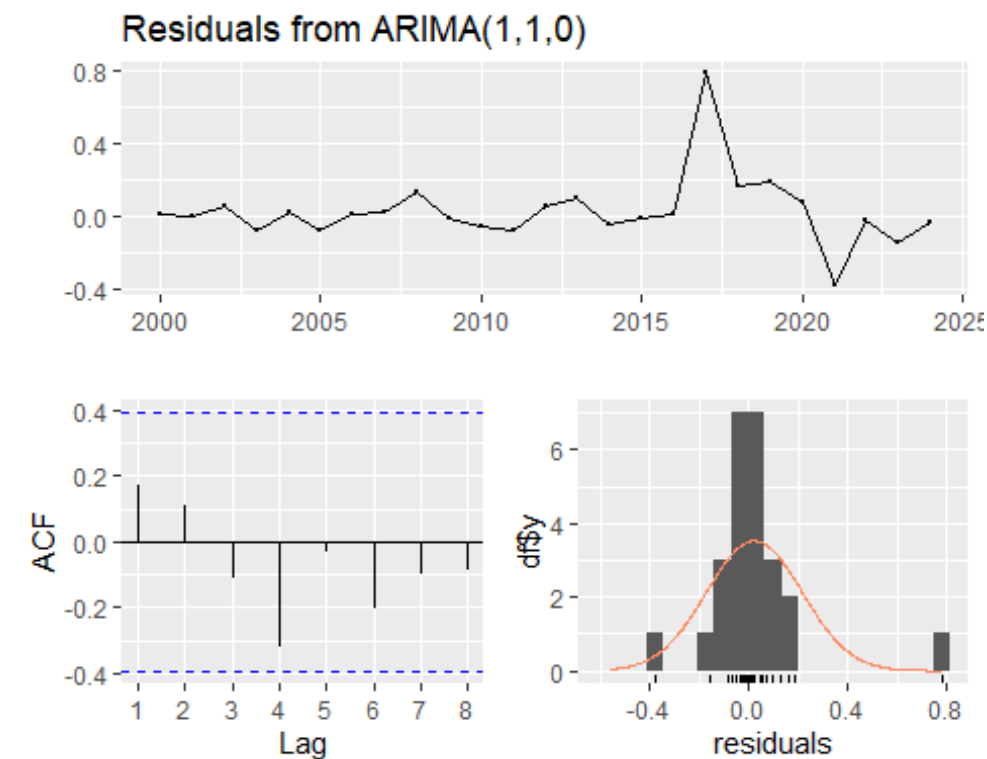
## Series: unemp_ts
## ARIMA(1,1,0)
##
## Coefficients:
##      ar1
##      0.7374
## s.e.  0.1250
##
## sigma^2 = 0.03938: log likelihood = 4.88
## AIC=-5.75   AICc=-5.18   BIC=-3.4
##
## Training set error measures:
```

```
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 0.02279687 0.1903484 0.1037104 0.7685899 2.799581 0.6285479
##           ACF1
## Training set 0.1740154
```

Model Diagnostics

Residual analysis

```
checkresiduals(auto_model)
```



```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,1,0)
## Q* = 4.9559, df = 4, p-value = 0.2919
##
## Model df: 1.   Total lags used: 5
```

Ljung-Box test for residual autocorrelation

```
Box.test(auto_model$residuals, lag = 10, type = "Ljung-Box")
```

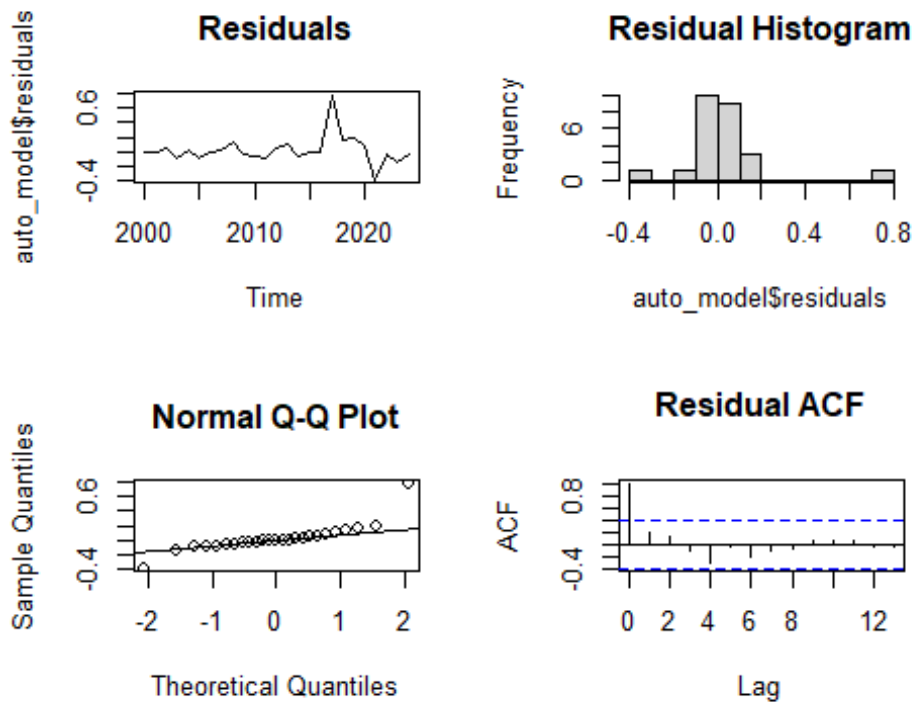
```
##  
## Box-Ljung test  
##  
## data: auto_model$residuals  
## X-squared = 7.3708, df = 10, p-value = 0.69
```

Normality test on residuals

```
shapiro.test(auto_model$residuals)  
  
##  
## Shapiro-Wilk normality test  
##  
## data: auto_model$residuals  
## W = 0.73652, p-value = 2.354e-05
```

Plot residuals

```
par(mfrow = c(2,2))  
plot(auto_model$residuals, type = "l", main = "Residuals")  
hist(auto_model$residuals, breaks = 10, main = "Residual Histogram")  
qqnorm(auto_model$residuals); qqline(auto_model$residuals)  
acf(auto_model$residuals, main = "Residual ACF")
```



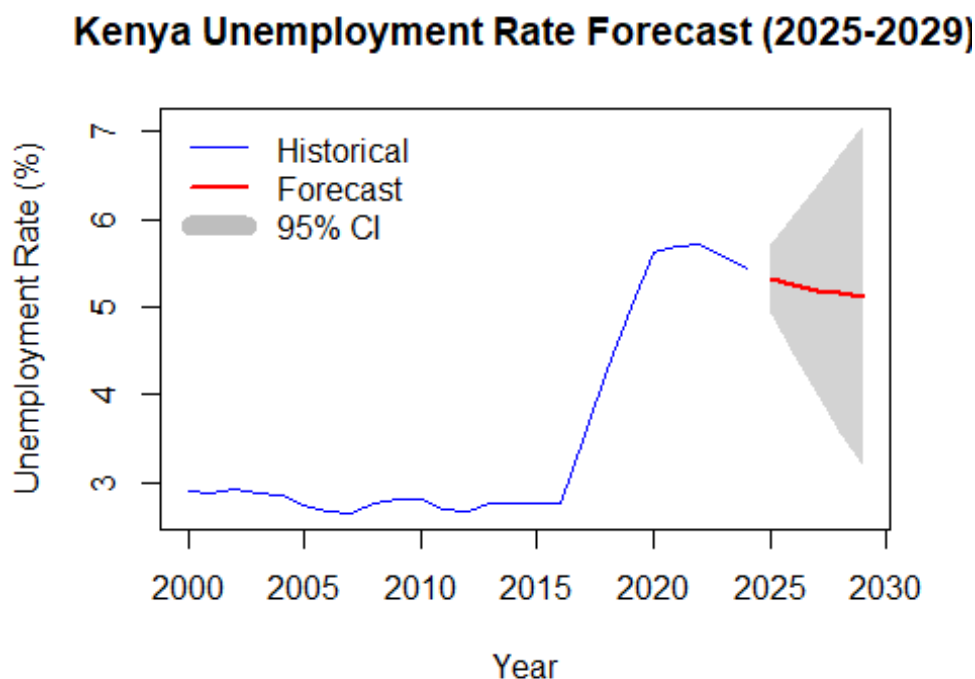
Forecasting and Visualization

Generate forecasts (5 years ahead)

```
forecast_horizon <- 5  
unemp_forecast <- forecast(auto_model, h = forecast_horizon)
```

Plot forecast with confidence intervals

```
plot(unemp_forecast,  
     include = 2000:2024,  
     main = "Kenya Unemployment Rate Forecast (2025-2029)",  
     xlab = "Year", ylab = "Unemployment Rate (%)",  
     col = "blue", shadecols = "lightgray")  
lines(2025:2029, unemp_forecast$mean, col = "red", lwd = 2)  
legend("topleft", legend = c("Historical", "Forecast", "95% CI"),  
      col = c("blue", "red", "gray"), lwd = c(1,2,10), bty = "n")
```



Extract forecast values

```
forecast_df <- data.frame(  
  Year = 2025:2029,  
  Forecast = as.numeric(unemp_forecast$mean),  
  Lo95 = as.numeric(unemp_forecast$lower[,2]),
```



```

    Hi95 = as.numeric(unemp_forecast$upper[,2])
  )
print(forecast_df)

##   Year Forecast      Lo95      Hi95
## 1 2025 5.326765 4.937807 5.715724
## 2 2026 5.250641 4.470924 6.030358
## 3 2027 5.194507 4.013319 6.375695
## 4 2028 5.153114 3.577196 6.729033
## 5 2029 5.122592 3.166831 7.078353

```

Project Summary

The ARIMA model reveals a concerning structural shift in Kenya's unemployment trajectory. After remaining relatively stable at 2.7-2.9% from 2000-2016, unemployment surged dramatically to approximately 5.6% by 2020-2022. The forecast indicates a modest declining trend, with rates projected to decrease gradually from 5.3% in 2025 to approximately 5.1% by 2029. However, the widening 95% confidence interval suggests increasing uncertainty in long-term predictions. While the slight downward trend is encouraging, unemployment rates remain significantly elevated compared to historical levels, indicating the need for sustained policy interventions to accelerate job creation and economic recovery.